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Metodi Supersimmetrici per la Meccanica Quantistica

Tesi di Laurea

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Abstract

In this thesis the formalism of Supersymmetry is applied to non-relativistic Quantum Mechanics and it is shown how new solvable potentials can be constructed from known ones. This formalism is then enriched with the concept of shape invariance, that leads to an algebraic ladder operator method for the solutions of certain potentials. Finally a Supersymmetry-inspired approach to the semi-classical WKB approximation is presented and its advantages with shape invariant potentials are stated. Some examples for each concept are provided.

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Introduction

The theoretical framework of Supersymmetry is based on the existence of a new kind of symmetry in which to each bosonic particle corresponds a new fermion and to each fermionic particle corresponds a new boson: these are called superpartner particles and their spin differs from their "regular" partner's by a half integer. More precisely, Supersymmetry was introduced in the context of Quantum Field Theories as an internal symmetry of a Lagrangian between fermionic and bosonic degrees of freedom, as a way to unify interactions (bosons) with matter (fermions). So far in nature none of these superpartners have been observed and so it is conjectured that Supersymmetry must be spontaneously broken and the scale of this breaking is at least of the order of 100 GeV.

From this context, researchers started to test supersymmetric theories in simpler theoretical frameworks, like that of non-relativistic Quantum Mechanics, finding insight on solvable potentials.

In the first chapter of this thesis the formalism and algebraic structure of Supersymmetric Quantum Mechanics is introduced and the properties of supersymmetric partner Hamiltonians are stated and proven; some examples are then provided and explicitely explored. The main references for this chapter are [1] and [2] for the definitions and properties of Supersymmetry and [3] for the Supersymmetric Algebra.

In the second chapter of this thesis the concept of shape invariant partner potentials is introduced with their properties: this definition is then used to construct a generalization of the ladder operators for the Quantum Harmonic Oscillator to a greater set of integrable 1-dimensional potentials; finally, some examples of potentials with complex analytical solutions are solved with the properties of shape invariance. The main reference for this chapter is again [1].

In the third and final chapter of this thesis a variation of the WKB quantization condition on bound states of supersymmetric partner potentials is derived for both broken and unbroken supersymmetry; it is then shown that this method yields the exact energy eigenvalues of the spectrum of shape invariant potentials. The references for this chapter are [4] for the general theory on the WKB method and [1], [5] for the supersymmetric approach.

Chapter 1

Supersymmetric Quantum Mechanics

Here the formalism of Hamiltonian 1-dimensional supersymmetric Quantum Mechanics is introduced, then its fundamental properties will be stated and proven; in the end, some applications to 1-dimensional and 3-dimensional potentials will be explored in detail.

1.1 First definitions

The fundamental structure of supersymmetric Quantum Mechanics is based on two operators:

$$\hat{A} = \frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x)$$
$$\hat{A}^\dagger = -\frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x).$$

that will be used to construct two Hamiltonians.

Definition 1.1. The real function $W(x)$ is called *superpotential*.

Definition 1.2. The self-adjoint operators

$$\hat{H}_1 := \hat{A}^\dagger \hat{A} \tag{1.1}$$
$$\hat{H}_2 := \hat{A}\hat{A}^\dagger \tag{1.2}$$

are called *supersymmetric partners Hamiltonians*, and it is trivial to show that

$$\hat{H}_1 = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_1(x)$$
$$\hat{H}_2 = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_2(x),$$

where

$$V_1(x) = W^2(x) - \frac{\hbar}{\sqrt{2m}}W'(x) \tag{1.3}$$
$$V_2(x) = W^2(x) + \frac{\hbar}{\sqrt{2m}}W'(x). \tag{1.4}$$

These definitions employ the typical language of supersymmetric theories: its fundamental algebraic structure will now be briefly introduced.

1.2 Hints at supersymmetric Algebra

Similarly to classical Hamiltonian mechanics, in Quantum Mechanics every symmetry is closely related to an operator commuting with the Hamiltonian of the system: Supersymmetry can also be defined in similar terms.

Given the operators defined earlier, consider the operators

$$\hat{Q} = \begin{pmatrix} \mathbb{O} & \mathbb{O} \\ \hat{A} & \mathbb{O} \end{pmatrix}, \hat{Q}^\dagger = \begin{pmatrix} \mathbb{O} & \hat{A}^\dagger \\ \mathbb{O} & \mathbb{O} \end{pmatrix}$$

and the Hamiltonian

$$\hat{H} := \left\{\hat{Q}, \hat{Q}^\dagger\right\} = \begin{pmatrix} \hat{A}^\dagger\hat{A} & \mathbb{O} \\ \mathbb{O} & \hat{A}\hat{A}^\dagger \end{pmatrix} = \begin{pmatrix} \hat{H}_1 & \mathbb{O} \\ \mathbb{O} & \hat{H}_2 \end{pmatrix},$$

The (vector) ground state of $\hat{H}$ is

$$\psi_0 = \begin{pmatrix} \psi_0^{(1)} \\ \psi_0^{(2)} \end{pmatrix},$$

where $\psi_0^{(i)}$ are the ground states of $\hat{H}_i$.

The definition of the Hamiltonian $\hat{H}$ is the first property of the supersymmetric Algebra, the others follow from the definitions of $\hat{Q}$ and $\hat{H}$:

$$\left[\hat{H}, \hat{Q}\right] = \left[\hat{H}, \hat{Q}^\dagger\right] = 0$$

and

$$\left\{\hat{Q}, \hat{Q}\right\} = \left\{\hat{Q}^\dagger, \hat{Q}^\dagger\right\} = 0.$$

Note The symmetry is given by the fact that $\hat{Q}$, $\hat{Q}^\dagger$ commute with the Hamiltonian of the system.

To introduce the concept of supersymmetry breaking, some properties of the partner Hamiltonians need to be considered.

1.3 Fundamental properties of supersymmetric partners

The most interesting properties of supersymmetric partner Hamiltonians lie in the relationships between their energy spectra and corresponding eigenstates.

Theorem 1.3. *Supersymmetric partner Hamiltonians admit only non-negative eigenvalues.*

Proof Consider the Hamiltonians $\hat{H}_1, \hat{H}_2$: $\forall\, |\psi\rangle$ the quantities

$$\langle\psi|\, \hat{H}_1\, |\psi\rangle = \langle\psi|\, (\hat{A}^\dagger\hat{A})\, |\psi\rangle = \left|\left|\hat{A}\, |\psi\rangle\right|\right|^2$$
$$\langle\psi|\, \hat{H}_2\, |\psi\rangle = \langle\psi|\, (\hat{A}\hat{A}^\dagger)\, |\psi\rangle = \left|\left|\hat{A}^\dagger\, |\psi\rangle\right|\right|^2$$

are non-negative, thus proving the statement. □

In addition to this interesting property, supersymmetric partner Hamiltonians' spectra are related, due to the fact that they are constructed with the same two operators: consider the eigenstate $\left|E^{(1)}\right\rangle$ for $\hat{H}_1$ with eigenvalue E, such that

$$\hat{H}_1 \left|E^{(1)}\right\rangle = E \left|E^{(1)}\right\rangle.$$

Applying the operator $\hat{A}$ to both sides yields

$$\hat{A}\hat{H}_1 \left|E^{(1)}\right\rangle = E\hat{A} \left|E^{(1)}\right\rangle.$$

Recalling (1.1) and (1.2), **if** $\hat{\mathbf{A}} \left|\mathbf{E^{(1)}}\right\rangle \neq \mathbf{0}$ (thus $E \neq 0$), this is equivalent to

$$\hat{A}\hat{A}^\dagger\hat{A} \left|E^{(1)}\right\rangle = E\hat{A} \left|E^{(1)}\right\rangle \Rightarrow \hat{H}_2\hat{A} \left|E^{(1)}\right\rangle = E\hat{A} \left|E^{(1)}\right\rangle.$$

This shows that the state

$$\left|E^{(2)}\right\rangle := \frac{1}{\sqrt{E}}\hat{A} \left|E^{(1)}\right\rangle \tag{1.5}$$

is a normalized eigenstate of $\hat{H}_2$ with eigenvalue E.

Similarly, it can be shown that for an eigenstate $\left|E^{(2)}\right\rangle$ of $\hat{H}_2$, **if** $\hat{\mathbf{A}}^\dagger \left|\mathbf{E^{(2)}}\right\rangle \neq \mathbf{0}$, the state

$$\left|E^{(1)}\right\rangle := \frac{1}{\sqrt{E}}\hat{A}^\dagger \left|E^{(2)}\right\rangle \tag{1.6}$$

is a normalized eigenstate of $\hat{H}_1$ with eigenvalue E.

If, on the other hand, $\hat{A}\left|E^{(1)}\right\rangle = 0$ or $\hat{A}^\dagger\left|E^{(2)}\right\rangle = 0$, the eigenvalue of both these eigenstates must be $E = 0$; to see the implications of this condition, it is useful to consider the energy spectra of $\hat{H}_1, \hat{H}_2$ to be discrete, and so from this point onwards the eigenfunctions and eigenvalues of the Hamiltonians will be denoted as

$$\hat{H}_i\psi_n^{(i)} = E_n^{(i)}\psi_n^{(i)}.$$

Consider now the case $\hat{A}\psi_n^{(1)} = 0$: this implies $\hat{H}_1\psi_n^{(1)} = 0$, i.e. that the wavefunction annihilated by $\hat{A}$ is the ground state of $\hat{H}_1$, with eigenvalue $E_0^{(1)} = 0$; this equality yields an analytic expression for the ground state of $\hat{H}_1$:

$$\hat{A}\psi_0^{(1)} = 0 \iff \frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x}\psi_0^{(1)}(x) = W(x)\psi_0^{(1)} \iff \frac{\mathrm{d}\psi_0^{(1)}}{\psi_0^{(1)}} = -\frac{\sqrt{2m}}{\hbar}W(x)$$

$$\psi_0^{(1)}(x) = N\exp\left[-\frac{\sqrt{2m}}{\hbar}\int_{x_0}^{x}\mathrm{d}yW(y)\right]. \tag{1.7}$$

Note For this to be a true ground state wavefunction, it must be continuous and normalizable: this is true if the integral function

$$f(x) = \int_{x_0}^{x}\mathrm{d}yW(y)$$

is continuous and if

$$\lim_{x\to\pm\infty}\frac{f(x)}{\ln(\pm x)}=+\infty:$$

this means that f must increase at $+\infty$ and decrease at $-\infty$, since $f(x)$ is the integral function of $W(x)$, this implies that

$$\begin{cases}0\leq\lim\limits_{x\to+\infty}W(x)\leq+\infty\\ -\infty\leq\lim\limits_{x\to-\infty}W(x)\leq 0.\end{cases}\tag{1.8}$$

If the ground state is known, $\hat{A}\psi_0^{(1)}=0$ provides an expression for the superpotential

$$W(x)=-\frac{\hbar}{\sqrt{2m}}\frac{\psi_0'^{(1)}(x)}{\psi_0^{(1)}(x)}.\tag{1.9}$$

These considerations about the ground state can be similarly derived using $\hat{H}_2$, from $\hat{A}^\dagger\psi_0^{(2)}=0$:

$$\psi_0^{(2)}(x)=N\exp\left[\frac{\sqrt{2m}}{\hbar}\int_{x_0}^{x}\mathrm{d}yW(y)\right],$$

$$\lim_{x\to\pm\infty}\frac{f(x)}{\ln(\pm x)}=-\infty\implies\begin{cases}-\infty\leq\lim\limits_{x\to+\infty}W(x)\leq 0\\ 0\leq\lim\limits_{x\to-\infty}W(x)\leq+\infty\end{cases},$$

$$W(x)=\frac{\hbar}{\sqrt{2m}}\frac{\psi_0'^{(2)}(x)}{\psi_0^{(2)}(x)}.$$

It is easy to show that these two conditions (and so two $E=0$ ground states) cannot cohexist:

Theorem 1.4. *Only one of the two ground states of partner Hamiltonians can have eigenvalue* $E_0^{(i)}=0$.

Proof (by contradiction) Suppose both ground states have $E_0^{(1)}=E_0^{(2)}=0$: by the previous calculations, their wavefunctions are

$$\begin{matrix}\psi_0^{(1)}=N\exp\left[-f(x)\right]\\ \psi_0^{(2)}=N\exp\left[f(x)\right]\end{matrix}\text{ with }f(x)=\frac{\sqrt{2m}}{\hbar}\int_{x_0}^{x}\mathrm{d}yW(y)$$

it is clear that these two wavefunctions cannot be simultaneously normalized, a contradiction. □

Note From this point, the superpotential $W(x)$ is chosen so that only the Hamiltonian $\hat{H}_1$ can possess a $E=0$ ground state.

Definition 1.5. Supersymmetry is **spontaneously broken** if and only if

$$\hat{A}\psi_0^{(1)} \neq 0.$$

Definition 1.6. Supersymmetry is **unbroken** if and only if

$$\hat{A}\psi_0^{(1)} = 0.$$

Note Since $\hat{A}\psi_0^{(1)} = 0$ is a necessary condition for (1.8), if the limits of $W(x)$ do not respect these conditions, Supersymmetry must be spontaneously broken.
If Supersymmetry is spontaneously broken, the limits of $W(x)$ can have any sign save those described in (1.8) and their counterpart implied by $\hat{A}\psi_0^{(2)} = 0$: this implies that they must have the same sign.

Another way to define Supersymmetry breaking is through the operators $\hat{Q}, \hat{Q}^\dagger$ defined in Section 1.2:

$$\hat{Q}^\dagger\psi_0 = 0 \iff \psi_0^{(2)} \equiv 0 \wedge \hat{A}\psi_0^{(1)} = 0 \iff \hat{Q}\psi_0 = 0,$$

so Supersymmetry is unbroken if and only if

$$\hat{Q}\psi_0 = \hat{Q}^\dagger\psi_0 = 0$$

which closely resembles the definition of spontaneous symmetry breaking in Quantum Mechanics.

For both broken and unbroken Supersymmetry, the peculiar property of supersymmetric partner Hamiltonians is expressed in equations (1.5) and (1.6): for discrete spectra, the ground state will always be denoted by $n = 0$, so the labels of the energy levels will be shifted in case of unbroken Supersymmetry

$$E_n^{(2)} = E_{n+1}^{(1)},\ \forall n = 0, 1, 2...$$

while the labels remain the same if supersymmetry is broken

$$E_n^{(2)} = E_n^{(1)} \neq 0,\ \forall n = 0, 1, 2...$$

The relationships between the eigenfunctions $\psi_n^{(i)}(x)$ with eigenvalues $E_n^{(i)} \neq 0$ are then

$$\psi_n^{(2)}(x) = \frac{1}{\sqrt{E_{n+1}^{(1)}}}\hat{A}\psi_{n+1}^{(1)}(x) \tag{1.10}$$

$$\psi_n^{(1)}(x) = \frac{1}{\sqrt{E_{n-1}^{(2)}}}\hat{A}^\dagger\psi_{n-1}^{(2)}(x) \tag{1.11}$$

This shows that knowing the full energy spectrum and eigenfunctions of one of the partner Hamiltonians is enough to analytically construct the solutions of the other Hamiltonian, except at most an $E = 0$ ground state: this is the fundamental concept of Supersymmetry that will be used in this thesis.

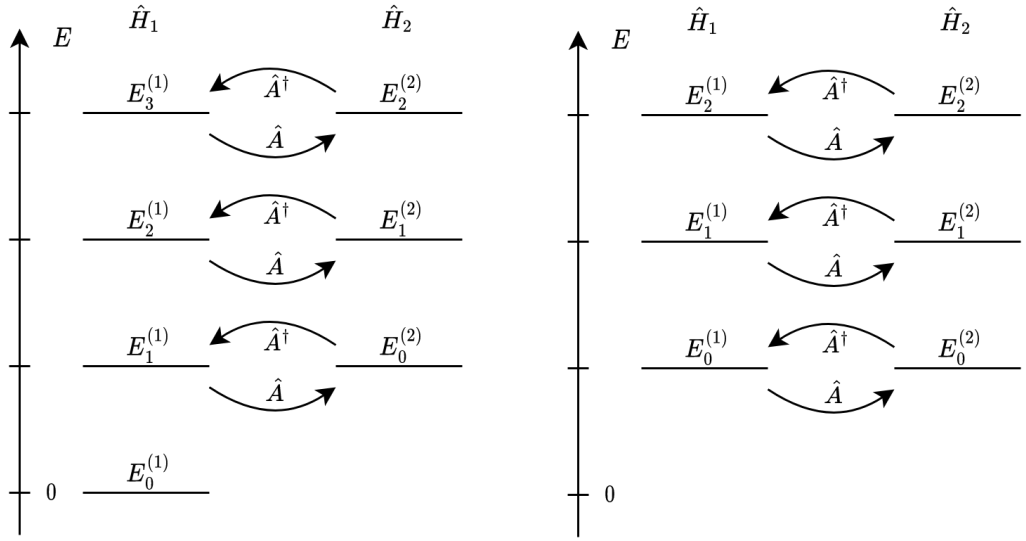


Figure 1.1: Diagram illustrating the energy levels of $\hat{H}_1$, $\hat{H}_2$ in case of unbroken and broken Supersymmetry (on the left and right, respectively) and the operators relating their respective eigenfunctions: unbroken Supersymmetry is characterized by the $E_0^{(1)} = 0$ eigenstate.

Note From this point onwards Supersymmetry is considered to be unbroken, unless otherwise specified.

1.4 Example: the infinite square well

Given a simply solvable potential $V_1(x)$, its partner potential $V_2(x)$ can be found and exactly solved: from this second potential a third ($V_3(x)$) can be derived and so on: in this section this method is demonstrated beginning from the infinite square well potential.

Consider the potential

$$V_1(x) = \begin{cases} -\dfrac{\pi^2\hbar^2}{2mL^2} & \text{if } 0 \leq x \leq L \\ \infty & \text{otherwise} \end{cases}.$$

This is the well known infinite square well potential, with an offset so that its ground state has $E_0^{(1)} = 0$: its energy eigenvalues and corresponding eigenstates are

$$\begin{aligned} E_n^{(1)} &= \frac{\pi^2\hbar^2}{2mL^2} n(n+2), \\ \psi_n^{(1)}(x) &= \sqrt{\frac{2}{L}} \sin\left(\frac{(n+1)\pi x}{L}\right) \end{aligned} \quad \text{with } n = 0, 1, 2, 3...$$

Since Supersymmetry is unbroken, the superpotential can be found using equation (1.9):

$$W(x) = -\frac{\hbar}{\sqrt{2m}} \frac{\psi_0'^{(1)}(x)}{\psi_0^{(1)}(x)} = -\frac{\hbar}{\sqrt{2m}} \frac{\frac{\pi}{L}\cos\left(\frac{\pi x}{L}\right)}{\sin\left(\frac{\pi x}{L}\right)} = -\frac{\hbar\pi}{L\sqrt{2m}} \cot\left(\frac{\pi x}{L}\right)$$

then, from equation (1.4), the partner potential to $V_1(x)$ is

$$V_2(x) = W^2(x) + \frac{\hbar}{\sqrt{2m}}W'(x) = \frac{\pi^2\hbar^2}{2mL^2}\left[\cot^2\left(\frac{\pi x}{L}\right) + \csc^2\left(\frac{\pi x}{L}\right)\right] = \\ = \frac{\pi^2\hbar^2}{2mL^2}\left[2\csc^2\left(\frac{\pi x}{L}\right) - 1\right].$$

Using the properties of superymmetric partner potentials, the energy spectrum of $\hat{H}_2$ is readily determined:

$$E_n^{(2)} = E_{n+1}^{(1)} = \frac{\pi^2\hbar^2}{2mL^2}(n+1)(n+3) \quad \text{with } n = 0, 1, 2, 3...$$

as said before, the $E = 0$ eigenstate exists only for $\hat{H}_1$.

The eigenfunctions of $\hat{H}_2$ can be analytically determined with equation (1.10): the ground state of $\hat{H}_2$ is

$$\psi_0^{(2)}(x) = \frac{1}{\sqrt{E_1^{(1)}}}\hat{A}\psi_1^{(1)} = \frac{1}{\sqrt{E_1^{(1)}}}\left[\frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x)\right]\psi_1^{(1)}(x) = -2\sqrt{\frac{2}{3L}}\sin^2\left(\frac{\pi x}{L}\right),$$

with eigenvalue $E_0^{(2)} = \frac{3\pi^2\hbar^2}{2mL^2}$.
The first excited state of $\hat{H}_2$ is

$$\psi_1^{(2)}(x) = \frac{1}{\sqrt{E_2^{(1)}}}\hat{A}\psi_2^{(1)} = -\frac{2}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{2\pi x}{L}\right),$$

with eigenvalue $E_1^{(2)} = \frac{8\pi^2\hbar^2}{2mL^2}$.

The complete derivation of these wavefunctions can be found in Appendix 4.4.

This process can be iterated: the potential $V_2(x)$ can be redefined so that its ground state energy is 0, a new superpotential $W_2(x)$ can be determined and from this arises a third potential $V_3(x)$, supersymmetric partner of $V_2(x) - E_0^{(2)}$, with the same energy spectrum, except for the $E = 0$ eigenvalue; using the new superpotential, the eigenfunctions of $\hat{H}_3$ can be analytically determined in a similar fashion as before.

This shows that, using the properties of supersymmetric partner potentials, an entire class of analytically solvable potentials can be constructed starting from an analytically solved problem.

1.5 Radial potentials

The formalism of Supersymmetry can be easily applied to 3-dimensional radial potentials: as is well known, if a potential $V(r, \vartheta, \varphi)$ is independent of the angular coordinates, the eigenstates of the Hamiltonian can be factored in

$$\psi_{nlm}(r, \vartheta, \varphi) = \frac{1}{r}R_{nl}(r)Y_{lm}(\vartheta, \varphi),$$

where Y_{lm} are the spherical harmonics and the reduced radial wavefunction $R_{nl}(r)$ satisfies

$$-\frac{\mathrm{d}^2 R_{nl}}{\mathrm{d}r^2} + \left[V(r) + \frac{l(l+1)}{r^2}\right] R_{nl}(r) = E R_{nl}(r),$$

in units where $\hbar = 2m = 1$.
This is a 1-dimensional Schrödinger equation with an effective potential that incorporates the angular momentum barrier.

1.5.1 Example: Oscillator potential

Consider the superpotential

$$W(r) = \frac{\omega r}{2} - \frac{l+1}{r} : \tag{1.12}$$

its associated partner potentials are

$$\begin{cases} V_1(r) = \dfrac{\omega^2 r^2}{4} + \dfrac{l(l+1)}{r^2} - \left(l + \dfrac{3}{2}\right)\omega \\ V_2(r) = \dfrac{\omega^2 r^2}{4} + \dfrac{(l+1)(l+2)}{r^2} - \left(l + \dfrac{1}{2}\right)\omega \end{cases} . \tag{1.13}$$

The potential $V_1(r)$ is the effective potential for the isotropic harmonic oscillator: its l'th partial wave has the $l+1$'th partial wave of the same potential as its supersymmetric partner. This radial potential will be used later as an example of the advantages of the supersymmetric approach.

1.6 Transmission and reflection coefficients

Another implication of the properties of supersymmetric Quantum Mechanics is the relationship between transmission and reflection coefficients of two partner potentials in scattering problems: before deriving it, it is necessary to establish the conditions on $W(x)$ that guarantee the existence of unbound eigenstates.

Consider the usual partner potentials $V_1(x)$ and $V_2(x)$, in units where $\hbar = 2m = 1$:

$$V_1(x) = W^2(x) - W'(x)$$
$$V_2(x) = W^2(x) + W'(x).$$

The corresponding Hamiltonians $\hat{H}_1$, $\hat{H}_2$ admit unbound eigenstates if and only if

$$\lim_{x\to\pm\infty} V_i(x) = L_\pm^{(i)} \in \mathbb{R},$$

and the only way for $V_i(x)$ to have finite limits at infinity is if

$$\lim_{x\to\pm\infty} W(x) = W_\pm \in \mathbb{R} \iff \lim_{x\to\pm\infty} V_i(x) = W_\pm^2.$$

Under this condition on $W(x)$, the partner Hamiltonians admit unbound eigenstates.

Consider a progressive plane wave of energy E scattering on the potential $V_i(x)$: the wavefunctions $\psi^{(i)}(x)$ will have limits

$$\begin{cases} \lim_{x\to-\infty} \psi^{(i)}(x) = e^{ikx} + R_i e^{-ikx} \\ \lim_{x\to+\infty} \psi^{(i)}(x) = T_i e^{ik'x} \end{cases}, \text{ with } \begin{array}{l} k = \sqrt{E - W_-^2} \\ k' = \sqrt{E - W_+^2} \end{array},$$

where R_i are the *reflection coefficients* and T_i the *transmission coefficients* of the two potentials.

Applying property (1.11) in the limit $x \to -\infty$ yields

$$\begin{aligned} e^{ikx} + R_1 e^{-ikx} &= N\left[-\frac{\mathrm{d}}{\mathrm{d}x}e^{ikx} + W_- e^{ikx} + R_2\left(-\frac{\mathrm{d}}{\mathrm{d}x}e^{-ikx} + W_- e^{-ikx}\right)\right] = \\ &= Ne^{ikx}(-ik + W_-) + Ne^{-ikx}(ik + W_-)R_2 : \end{aligned}$$

equating the coefficients of the exponentials

$$\begin{cases} R_1 = N(W_- + ik)R_2 \\ N = \dfrac{1}{W_- - ik} \end{cases} \iff R_1 = \frac{W_- + ik}{W_- - ik}R_2. \tag{1.14}$$

Applying the same property to the limit $x \to +\infty$ gives

$$T_1 e^{ik'x} = N\left(-\frac{\mathrm{d}}{\mathrm{d}x} + W_+\right)e^{ik'x}T_2 = N(W_+ - ik')T_2,$$

and it follows that

$$T_1 = \frac{W_+ - ik'}{W_- - ik}T_2. \tag{1.15}$$

These two relations imply

$$\begin{aligned} |R_1|^2 &= |R_2|^2 \\ |T_1|^2 &= |T_2|^2, \end{aligned}$$

which means the transmission and reflection probabilities are the same for both partner potentials: this result is important, especially in more than 1 dimension, as it relates the S-matrices of different scattering problems.

1.7 Example: Reflectionless potentials

In this section the relationship between the transmission and reflection coefficients of partner Hamiltonians will be used to construct a family of algebraically solvable potentials, starting from the free particle, which has reflection coefficient $R = 0$.

Consider the superpotential

$$W(x; B) = B\tanh(x)$$

and its partner potentials

$$\begin{aligned} V_1(x; B) &= B^2 - B(B+1)\cosh^{-2}(x) \\ V_2(x; B) &= B^2 - B(B-1)\cosh^{-2}(x). \end{aligned}$$

Both potentials have finite limits, because

$$\lim_{x\to\pm\infty} W(x;B) = \pm B$$
$$\lim_{x\to\pm\infty} V_i(x;B) = B^2,$$

so they always admit unbound eigenstates with $E \geq B^2$.

Consider $B = 1$: the superpotential and potentials are

$$W(x;B=1) = \tanh(x)$$
$$\begin{cases} V_1(x;B=1) = 1 - 2\cosh^{-2}(x) \\ V_2(x;B=1) = 1 \end{cases}$$

It is clear that $\hat{H}_2(B=1)$ is the Hamiltonian of a free particle, so its transmission and reflection coefficients are

$$\begin{cases} T_2(k;B=1) = 1 \\ R_2(k;B=1) = 0. \end{cases}$$

Then, from equations (1.14) and (1.15), it follows that the coefficients for H_1 are

$$\begin{cases} T_1(k;B=1) = \dfrac{1-ik}{-1-ik} \\ R_1(k;B=1) = 0 \end{cases}.$$

Using (1.7), the ground state wavefunction is

$$\psi_0^{(1)}(x;B=1) = N\exp\left[-\int_{x_0}^{x} \mathrm{d}y\,\tanh(y)\right] = N\cosh^{-1}(x)$$

and it is normalizable: from the properties discussed before, this implies that Supersymmetry is unbroken, so $\hat{H}_1(B=1)$ admits exactly one bound state with $E_0^{(1)}(B=1) = 0$ and a continuum of unbound states with $E^{(1)}(B=1) \geq 1$.

Consider now $B = 2$, the two potentials are

$$\begin{cases} V_1(x;B=2) = 4 - 6\cosh^{-2}(x) \\ V_2(x;B=2) = 4 - 2\cosh^{-2}(x) \end{cases}$$

and it is clear that $V_2(x;B=2)$ differs from $V_1(x;B=1)$ just by a constant, so its transmission and reflection coefficients must be the same (and their eigenfunctions are proportional to one another); using once again equations (1.14) and (1.15), the transmission coefficient of $\hat{H}_1(B=2)$ is

$$T_1(k;B=2) = \frac{2-ik}{-2-ik}\frac{1-ik}{-1-ik}$$

and its ground state wavefunction is

$$\psi_0^{(1)}(B=2) = N\cosh^{-2}(x)$$

so Supersymmetry must again be unbroken and $E_0^{(1)}(B=2)=0$; it is also clear that $\hat{H}_2(B=2)$ admits a bound state with $E_0^{(2)}(B=2)=3$ (its ground state), and so does $\hat{H}_1(B=2)$: the corresponding eigenfunction is

$$\psi_1^{(1)}(x;B=2)=\hat{A}^\dagger(B=2)\psi_0^{(2)}(x;B=2)=$$
$$=\hat{A}^\dagger(B=2)\psi_0^{(1)}(x;B=1)=N\tanh(x)\cosh^{-1}(x),$$

where the first equality is due to the properties of partner Hamiltonians and the second is due to the fact that $V_2(B=2)=V_1(B=1)+3$.

This process can be iterated: $\forall B=p\in\mathbb{N}$, the potentials $V_2(x;p)$ are reflectionless (as they differ from $V_1(x;p-1)$ just by a constant and so on), and so are $V_1(x;p)$, as they are partner potentials; their transmission coefficients are

$$T_1(k;p)=\frac{(p-ik)(p-1-ik)...(1-ik)}{(-p-ik)(-p+1-ik)...(-1-ik)}=\frac{\Gamma(-p-ik)\Gamma(p+1-ik)}{\Gamma(-ik)\Gamma(1-ik)},$$

their ground state wavefunctions are

$$\psi_0^{(1)}(x;p)=N\exp\left[-p\int_{x_0}^{x}\mathrm{d}y\tanh(y)\right]=N\cosh^{-p}(x),$$

the first excited state wavefunctions are

$$\psi_1^{(1)}(x;p)=\hat{A}^\dagger(p)\psi_0^{(1)}(x;p-1)=N\tanh(x)\cosh^{1-p}(x)$$

and finally the q'th excited state ($0<q<p$, as they admit p bound states) can be determined from the $q-1$'th as

$$\psi_q^{(1)}(x;p)=\hat{A}^\dagger(p)\psi_{q-1}^{(2)}(x;p)=\hat{A}^\dagger(p)\psi_{q-1}^{(1)}(x;p-1),$$

as can be deduced from Figure 1.2.

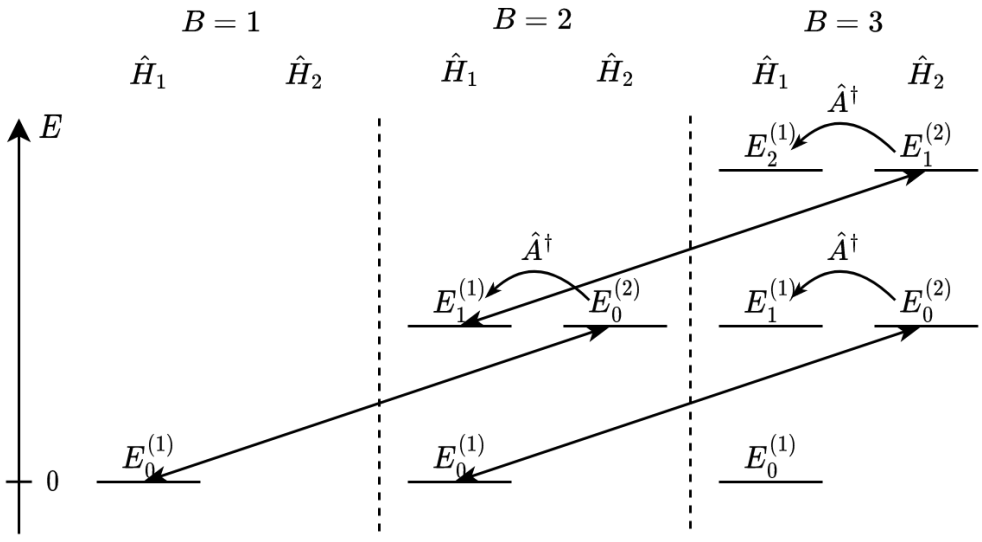


Figure 1.2: Diagram for the construction of the bound excited states of $\hat{H}_1(B\leq 3)$: the equality of the wavefunctions (save normalization) is depicted as a double arrow, while the use of the operator $\hat{A}^\dagger$ with a single arrow. The distribution of the levels is not to scale.

Using the properties of Supersymmetry, it was possible to construct a family of reflectionless potentials and easily compute their transmission coefficients and bound state wavefunctions, as they are all related to the free particle Hamiltonian.

In this example a more fundamental underlying property that relates the Hamiltonians $\hat{H}_i(B)$ to one another, called *shape invariance*, is at work: it will be discussed more thoroughly in the next chapter.

Chapter 2

Shape Invariance

This chapter will introduce a stricter condition on supersymmetric partner potentials, called *shape invariance*: the intuition behind this concept is the fact that certain potentials resemble one another and differ only in the choice of fixed parameters and/or constant energy offsets.

Here the formalization is presented and it is proven that this condition is a sufficient hypothesis to devise a generalization of the ladder operators, normally employed in the solution of the quantum harmonic oscillator; finally, some applications will be discussed in detail.

2.1 Definition and fundamental properties

Definition 2.1. Two supersymmetric partner potentials,

$$V_1(x;a_1) = W'(x;a_1) - W^2(x;a_1)$$
$$V_2(x;a_1) = W'(x;a_1) + W^2(x;a_1),$$

dependent from a set of parameters a_1, are called *shape invariant* if and only if

$$V_2(x;a_1) = V_1(x;a_2) + R(a_1), \tag{2.1}$$

where $a_2 = f(a_1)$ and $R(a_1)$ does not depend on x.

From this definition follows the fundamental property of shape invariant potentials: adding the kinetic operator to equation (2.1) gives

$$\hat{H}_2(a_1) = \hat{H}_1(a_2) + R(a_1) \longrightarrow \hat{H}_1(a_2) = \hat{H}_2(a_1) - R(a_1).$$

Since the Hamiltonians differ by a constant, their eigenfunctions must be proportional to one another and the corresponding energy eigenvalues must differ only by the constant $R(a_1)$: if Supersymmetry is unbroken, the ground state energy for $\hat{H}_2(a_1)$ is then known to be

$$E_0^{(2)}(a_1) = E_0^{(1)}(a_2) + R(a_1) = R(a_1)$$

and the ground states of $\hat{H}_1(a_2)$ and $\hat{H}_2(a_1)$ are proportional

$$\psi_0^{(1)}(x;a_2) = \psi_0^{(2)}(x;a_1) = N \exp\left[-\int_{x_0}^{x} \mathrm{d}y W(y;a_2)\right].$$

From the properties of Supersymmetry follows that

$$\psi_1^{(1)}(x;a_1) = \hat{A}^\dagger(a_1)\psi_0^{(2)}(x;a_1) \Rightarrow \psi_1^{(1)}(x;a_1) = \hat{A}^\dagger(a_1)\psi_0^{(1)}(x;a_2).$$

Since the function f that relates $a_2 = f(a_1)$ is known, the shape invariance condition implies a relationship between the ground state wavefunction and the first excited state! This process can be generalized to construct *all* bound states of $\hat{H}_1(a_1)$.

2.2 Generalized ladder operator method

To generalize the relationship found in the previous section, a family of Hamiltonians must be constructed from two supersymmetric partner potentials and the function f, which completely determines the shape invariance: starting from the partner potentials $V_1(x;a_1)$ and $V_2(x;a_1)$, consider the family of potentials

$$V_s(x;a_1) := V_1(x;a_s) + \sum_{k=1}^{s-1} R(a_k), \text{with } s = 1, 2... \tag{2.2}$$

where

$$a_s = f^{s-1}(a_1).$$

Theorem 2.2. *The potentials $V_s(x;a_1)$ and $V_{s+1}(x;a_1)$ are shape invariant potentials $\forall s = 1, 2...$*

Proof The base case $s = 1$ follows from definition 2.1.
For $s > 1$, let $b = a_s$: the potential $V_{s+1}(x;a_1)$ can be expressed as

$$V_{s+1}(x;a_1) = V_1(x;f(b)) + \sum_{k=1}^{s-1} R(a_k) - R(b),$$

using equation (2.1) with $a_1 := b$ yields

$$V_{s+1}(x;a_1) = V_2(x;b) + \sum_{k=1}^{s-1} R(a_k).$$

Since

$$V_s(x;a_1) = V_1(x;b) + \sum_{k=1}^{s-1} R(a_k),$$

$V_s(x;a_1)$ and $V_{s+1}(x;a_1)$ are respectively equal to $V_1(x;a_1)$ and $V_2(x;a_1)$, save a constant offset, so they are in turn supersymmetric partners and shape invariant potentials. □

Adding the kinetic operator provides a family of Hamiltonians

$$\hat{H}_s(a_1) = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V_1(x;a_s) + \sum_{k=1}^{s-1} R(a_k),$$

in which $\hat{H}_s$ and $\hat{H}_{s+1}$ are supersymmetric partners.
They can also be expressed in terms of $\hat{H}_1$ and $\hat{H}_2$:

$$\hat{H}_s(a_1) = \hat{H}_1(a_s) + \sum_{k=1}^{s-1} R(a_k)$$

$$\hat{H}_{s+1}(a_1) = \hat{H}_2(a_s) + \sum_{k=1}^{s-1} R(a_k).$$

Ignoring normalizations, this implies that the operators

$$\hat{A}_s := \hat{A}(a_s)$$
$$\hat{A}_s^\dagger := \hat{A}^\dagger(a_s)$$

relate their eigenfunctions just like $\hat{A}(a_1), \hat{A}^\dagger(a_1)$ relate those of $\hat{H}_1(a_1), \hat{H}_2(a_1)$.

If $\hat{H}_1$ admits p bound states, then $\hat{H}_s$ admits $p-s$ bound states, so it is clear that this process can be iterated at most $s = p-1$ times.

Using this method, the energy eigenvalues for $V_1(x; a_1)$ can be recursively determined: the ground state energy, if Supersymmetry is unbroken, is

$$E_0^{(1)}(a_1) = 0,$$

the first excited state's is

$$E_1^{(1)}(a_1) = E_0^{(2)}(a_1) = E_0^{(1)}(a_2) + R(a_1) = R(a_1),$$

where the first equality follows from Supersymmetry and the second from the shape invariance; the second excited state's is

$$E_2^{(1)}(a_1) = E_1^{(2)}(a_1) = E_0^{(3)}(a_1) = E_0^{(2)}(a_2) + R(a_1) =$$
$$= E_0^{(1)}(a_3) + R(a_1) + R(a_2) = R(a_1) + R(a_2).$$

Iterating this process shows that the n'th excited state (with $0 < n \leq p$) has energy

$$E_n^{(1)}(a_1) = \sum_{k=1}^{n} R(a_k), \tag{2.3}$$

and, since $\hat{H}_s(a_1)$ is a family of supersymmetric partners, this is also the ground state energy for $\hat{H}_{n+1}(a_1)$:

$$E_0^{(n+1)}(a_1) = E_n^{(1)}(a_1).$$

This method also provides an algebraic expression for the energy eigenfunctions of $V_1(x; a_1)$: since Supersymmetry is unbroken, the ground state wavefunction is

$$\psi_0^{(1)}(x; a_1) = N \exp\left[-\int_{x_0}^{x} \mathrm{d}y W(y; a_1)\right].$$

Using, respectively, Supersymmetry and shape invariance and ignoring normalization, the first excited state's wavefunction is

$$\psi_1^{(1)}(x; a_1) = \hat{A}_1^\dagger \psi_0^{(2)}(x; a_1) = \hat{A}_1^\dagger \psi_0^{(1)}(x; a_2)$$

and the second excited state's is

$$\psi_2^{(1)}(x; a_1) = \hat{A}_1^\dagger \psi_1^{(2)}(x; a_1) = \hat{A}_1^\dagger \hat{A}_2^\dagger \psi_0^{(3)}(x; a_1).$$

Using the shape invariance conditions

$$\psi_0^{(3)}(x; a_1) = \psi_0^{(2)}(x; a_2) = \psi_0^{(1)}(x; a_3)$$

yields

$$\psi_2^{(1)}(x;a_1) = \hat{A}_1^\dagger \hat{A}_2^\dagger \psi_0^{(1)}(x;a_3).$$

Iterating this line of reasoning, the n'th excited state's wavefunction is

$$\psi_n^{(1)}(x;a_1) = \hat{A}_1^\dagger ... \hat{A}_n^\dagger \psi_0^{(1)}(x;a_{n+1}), \tag{2.4}$$

or

$$\psi_n^{(1)}(x;a_1) = \hat{A}_1^\dagger \psi_{n-1}^{(1)}(x;a_2), \tag{2.5}$$

which effectively is a generalization of the ladder operators for the quantum harmonic oscillator, since the function f that relates the parameters is known.

2.3 Example: Reflectionless potentials

As stated at the end of 1.7, the family of reflectionless potentials associated with the partner potentials

$$\begin{cases} V_1(x;B) = B^2 - B(B+1)\cosh^{-2}(x) \\ V_2(x;B) = B^2 - B(B-1)\cosh^{-2}(x) \end{cases}, \text{ with } W(x;B) = B\tanh(x) \tag{2.6}$$

is also a family of shape invariant potentials: in this section the same results will be found using this property, in addition to the full energy spectra of the Hamiltonians.

Consider the potentials (2.6): it is easy to show that

$$V_2(x;B) = V_1(x;B-1) + B^2 - (B-1)^2,$$

so V_1 and V_2 are shape invariant potentials with

$$\begin{aligned} &a_1 := B, \\ &a_2 = f(a_1 = B) := B - 1, \\ &R(a_1 = B) := B^2 - (B-1)^2 = a_1^2 - a_2^2. \end{aligned}$$

Since f is a simple translation, a_n is readily computed to be

$$a_n = f^{n-1}(a_1 = B) = B - n + 1,$$

and so is $R(a_k)$:

$$R(a_k) = a_k^2 - a_{k+1}^2.$$

With these quantities, it is easy to apply the properties found in the previous section: Supersymmetry is unbroken, since the ground state wavefunction (1.7)

$$\psi_0^{(1)}(x;B) = N\exp\left[-B\ln\left(\cosh(x)\right)\right] = N\cosh^{-B}(x)$$

is normalizable, so the ground state energy must be $E_0^{(1)}(B) = 0$.

The other energy eigenvalues are immediately determined from equation (2.3):

$$E_n^{(1)}(B) = \sum_{k=1}^{n} R(a_k) = a_1^2 - a_2^2 + a_2^2 - ... - a_{n+1}^2 = a_1^2 - a_{n+1}^2 = B^2 - (B-n)^2.$$

These eigenvalues could not be determined as easily without the shape invariance condition.

To determine the eigenfunctions, consider first the operator $\hat{A}_s^\dagger$ earlier defined

$$\hat{A}_s^\dagger = \hat{A}^\dagger(a_s) = -\frac{\mathrm{d}}{\mathrm{d}x} + W(x; a_s) = -\frac{\mathrm{d}}{\mathrm{d}x} + (B - s + 1)\tanh(x).$$

The ground state eigenfunction was already determined to be

$$\psi_0^{(1)}(x; B) = N \cosh^{-B}(x),$$

so, using equation (2.5), the first excited state's wavefunction is

$$\psi_1^{(1)}(x; B) = \left[-\frac{\mathrm{d}}{\mathrm{d}x} + B\tanh(x)\right]\cosh^{1-B}(x) = N\tanh(x)\cosh^{1-B}(x),$$

the second's is

$$\psi_2^{(1)}(x; B) = \left[-\frac{\mathrm{d}}{\mathrm{d}x} + B\tanh(x)\right]\tanh(x)\cosh^{2-B}(x)$$

and so on.

2.4 Example: Radial Oscillator potential

Consider the radial superpotential (1.12)

$$W(r; l; \omega) = \frac{\omega r}{2} - \frac{l+1}{r}$$

and its associated partner potentials

$$\begin{cases} V_1(r; l; \omega) = \dfrac{\omega^2 r^2}{4} + \dfrac{l(l+1)}{r^2} - \left(l + \dfrac{3}{2}\right)\omega \\ V_2(r; l; \omega) = \dfrac{\omega^2 r^2}{4} + \dfrac{(l+1)(l+2)}{r^2} - \left(l + \dfrac{1}{2}\right)\omega \end{cases} :$$

it is easy to show that they are also shape invariant potentials, as

$$V_2(r; l; \omega) = V_1(r; l+1; \omega) + 2\omega,$$

in this case

$$\begin{aligned} a_1 &:= l, \\ a_2 &= f(a_1 = l) := l + 1. \end{aligned}$$

As in the first example, the function f is just a translation, so

$$\begin{aligned} a_k &= f^{k-1}(a_1) = l + k - 1, \\ R(a_k) &:= 2\omega \quad \forall k = 1, 2... \end{aligned}$$

As always, its ground state radial wavefunction can be determined from (1.7):

$$R_0^{(1)}(r; l; \omega) = N \exp\left[-\int_{r_0>0}^{r} \mathrm{d}\rho \frac{\omega\rho}{2} - \frac{l+1}{\rho}\right] = N r^{l+1} \exp\left[-\frac{\omega}{4} r^2\right],$$

so the $E_0^{(1)}(l;\omega) = 0$ eigenfunctions are

$$\psi_{0lm}^{(1)}(r,\vartheta,\varphi) = \mathcal{N} r^l \exp\left[-\frac{\omega}{4}r^2\right] Y_{lm}(\vartheta,\varphi).$$

Since this wavefunction is normalizable, Supersymmetry is unbroken, so $E_0^{(1)}(l;\omega) = 0$ and, using (2.3), the other energy eigenvalues are immediately determined to be

$$E_n^{(1)}(l;\omega) = \sum_{k=1}^{n} 2\omega = 2n\omega.$$

The excited states' wavefunctions can be determined using equation (2.5) recursively:

$$R_1^{(1)}(r;l;\omega) = \hat{A}^\dagger(l;\omega) R_0^{(1)}(r;l+1;\omega) = N\left(-\frac{\mathrm{d}}{\mathrm{d}r} + \frac{\omega}{2}r - \frac{l+1}{r}\right) r^{l+2} \exp\left[-\frac{\omega}{4}r^2\right],$$

which yields

$$R_1^{(1)}(r;l;\omega) = N\left(\omega r^2 - 2l - 3\right) r^{l+1} \exp\left[-\frac{\omega}{4}r^2\right],$$

so the $E_1^{(1)}(l;\omega)$ eigenfunctions are

$$\psi_{1lm}(r,\vartheta,\varphi) = \mathcal{N}(\omega r^2 - 2l - 3) r^l \exp\left[-\frac{\omega}{4}r^2\right] Y_{lm}(\vartheta,\varphi).$$

The purpose of this example is to show that a supersymmetric approach to a familiar problem can result in an elegant and brief algebraic solution.

Chapter 3

Supersymmetric WKB

The Wentzel–Kramers–Brillouin (WKB) approximation is a well known method of finding asymptotic series in powers of $\hbar$ for the energy eigenvalues and eigenfunctions of Schrödinger's equation: for bound states in a generalized potential well $V(x)$ which obeys the WKB condition

$$\left|\frac{\hbar p'(x)}{p^2}\right| = \left|\frac{\hbar m \frac{\partial V}{\partial x}}{p^3}\right| << 1, \text{ where } p(x) := \sqrt{2m\left[E - V(x)\right]}$$

i.e. slowly varying potentials and high momentum, the first semi-classical correction yields a quantization condition for the first order of the energy eigenvalues

$$\int_{x_1}^{x_2} \mathrm{d}y \sqrt{2m\left[E_n - V(y)\right]} = \left(n + \frac{1}{2}\right)\hbar\pi, \tag{3.1}$$

with $V(x_1) = V(x_2) = E_n$. It will be shown that this condition can yield more precise results using a supersymmetric approach called the SWKB approximation.

Note In this chapter the $\hbar$ and m dependencies of the potentials will be explicit, since they are necessary for the WKB approximation.

3.1 Lowest order SWKB: unbroken Supersymmetry

Consider two supersymmetric partner potentials

$$\begin{cases} V_1(x) = W^2(x) - \dfrac{\hbar}{\sqrt{2m}}W'(x) \\ V_2(x) = W^2(x) + \dfrac{\hbar}{\sqrt{2m}}W'(x) \end{cases}$$

so that, formally, $W(x) = O(\hbar^0)$ and $\hbar W'(x) = O(\hbar)$: the condition (3.1) for V_1 is

$$\int_{x_1}^{x_2} \mathrm{d}y \sqrt{2m\left[E_n^{(1)} - W^2(y) + \frac{\hbar}{\sqrt{2m}}W'(y)\right]} = \left(n + \frac{1}{2}\right)\hbar\pi;$$

expanding the integrand in powers of $\hbar$ yields

$$\sqrt{2m\left(E_n^{(1)} - W^2(y)\right)} + \frac{\hbar}{2}\frac{W'(y)}{\sqrt{E - W^2(y)}} + O(\hbar^2),$$

the condition $V_1(x_1) = V_1(x_2) = E_n^{(1)}$ must be truncated at $O(\hbar^0)$ and so becomes

$$W^2(a) = W^2(b) = E_n^{(1)}, \text{ where } x_1 = a + O(\hbar),\ x_2 = b + O(\hbar).$$

The condition (3.1) becomes

$$\int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(1)} - W^2(y)\right)} + \frac{\hbar}{2}\int_a^b \mathrm{d}y \frac{W'(y)}{\sqrt{E_n^{(1)} - W^2(y)}} = \left(n + \frac{1}{2}\right)\hbar\pi,$$

the second integral is elementar and its value is

$$\arcsin\left(\frac{W(y)}{\sqrt{E_n^{(1)}}}\right)\Bigg|_a^b.$$

Supposing Supersymmetry is unbroken, conditions (1.8) imply

$$-W(a) = W(b) = \sqrt{E_n^{(1)}}$$

that is

$$\arcsin\left(\frac{W(y)}{\sqrt{E_n^{(1)}}}\right)\Bigg|_a^b = \pi.$$

So, with the hypothesis of unbroken Supersymmetry, the first order SWKB quantization condition on $V_1(x)$ is

$$\int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(1)} - W^2(y)\right)} = n\hbar\pi. \tag{3.2}$$

The calculations for $V_2(x)$ are exactly the same, save a negative sign before the $O(\hbar)$ term: its SWKB quantization condition is

$$\int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(2)} - W^2(y)\right)} = (n+1)\hbar\pi. \tag{3.3}$$

The advantages of this method are immediately clear: since the integrand is strictly positive $\forall y \in [a,b]$, the quantization condition of V_1 with $n = 0$ implies that the extrema of integration must coincide, so

$$W(a) = -W(b) = -W(a) = 0 = \sqrt{E_0^{(1)}}:$$

the ground state eigenvalue is exactly known to be $E_0^{(1)} = 0$ just from the first $\hbar$ order; moreover, these approximations for the eigenvalues of the partner potentials respect the fundamental property of supersymmetric partner potentials $E_{n+1}^{(1)} = E_n^{(2)}$.

Note The regular WKB approximation yields better results the greater is n, but this supersymmetric method is also exact for the ground state energy: given this other constraint, it is natural to think that the SWKB method will be, in general, a better approximation in the case of unbroken Supersymmetry.

3.2 Lowest order SWKB: broken Supersymmetry

The derivation of the new quantization condition for broken supersymmetry is similar to the one for unbroken supersymmetry, with the key difference being that at the $O(\hbar^0)$ turning points the superpotential must have the same sign, as discussed after the definition of unbroken Supersymmetry:

$$W(a) = W(b) = \sqrt{E_n^{(1)}}$$

and so the $O(\hbar)$ integral vanishes for both V_1 and V_2, yielding

$$\int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(i)} - W^2(y)\right)} = \left(n + \frac{1}{2}\right)\hbar\pi$$

Note The reason why the SWKB method yields better results than regular WKB for both unbroken and broken supersymmetry lies in the fact that the factorization of supersymmetric partner Hamiltonians uncovers a term in the potential that is $O(\hbar^1)$, that can be taken out of the quantization condition integral and absorbed in the constant of the right hand side of the regular WKB condition. For further discussion on broken supersymmetry, see [6], as cited in [1]

3.3 SWKB for Shape Invariant Potentials

This method is especially powerful for shape invariant potentials: consider the family of potentials (2.2)

$$V_s(x; a_1) = V_1(x; a_s) + \sum_{k=1}^{s-1} R(a_k) :$$

the SWKB condition (3.2) for the Hamiltonian $\hat{H}_1$ has the form

$$\int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(1)} - W^2(y, a_s)\right)} = \int_a^b \mathrm{d}y \sqrt{2m\left(E_n^{(s)} - \sum_{k=1}^{s-1} R(a_k) - W^2(x; a_s)\right)} = n\hbar\pi.$$

If supersymmetry is unbroken, as argued before, the quantization condition is exact for the ground state energy

$$E_0^{(1)} = 0 \iff E_0^{(s)} = \sum_{k=1}^{s-1} R(a_k) :$$

since all the potentials are shape invariant, this implies that the ground state energy *for each Hamiltonian* $\hat{H}_s$ is exact, moreover, for the properties discussed in the previous chapter

$$E_0^{(s+1)} = E_s^{(1)}$$

and so **all** the energy eigenvalues of $\hat{H}_1$ are exactly determined by the SWKB quantization condition: this is a remarkable result, as the class of shape invariant potentials is bigger than the class of known analytically solvable potentials.

Appendix

4.4 Infinte square well

Here are shown the calculations for the ground state wavefunction of the potential V_2 determined in Section 1.4:

$$\psi_0^{(2)}(x) = \frac{1}{\sqrt{E_1^{(1)}}}\hat{A}\psi_1^{(1)} = \frac{1}{\sqrt{E_1}}\left[\frac{\hbar}{\sqrt{2m}}\frac{\mathrm{d}}{\mathrm{d}x} + W(x)\right]\psi_1^{(1)}(x) =$$
$$= \sqrt{\frac{2mL^2}{\pi^2\hbar^2 \cdot 1(1+2)}}\frac{\hbar}{\sqrt{2m}}\left[\frac{\mathrm{d}}{\mathrm{d}x} - \frac{\pi}{L}\cot\left(\frac{\pi x}{L}\right)\right]\sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right) =$$
$$= \sqrt{\frac{2}{3L}}\left[2\cos\left(\frac{2\pi x}{L}\right) - \frac{\cos\left(\frac{\pi x}{L}\right)}{\sin\left(\frac{\pi x}{L}\right)}2\cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi x}{L}\right)\right] = -2\sqrt{\frac{2}{3L}}\sin^2\left(\frac{\pi x}{L}\right)$$

The first excited state is

$$\psi_1^{(2)}(x) = \frac{1}{\sqrt{E_2^{(1)}}}\hat{A}\psi_2^{(1)}$$

letting $\alpha := \frac{\pi x}{L}$ this becomes

$$\psi_1^{(2)}(x) = \sqrt{\frac{2mL^2}{\pi^2\hbar^2 \cdot 2(2+2)}}\frac{\hbar}{\sqrt{2m}}\left[\frac{\pi}{L}\frac{\mathrm{d}}{\mathrm{d}\alpha} - \frac{\pi}{L}\cot(\alpha)\right]\sqrt{\frac{2}{L}}\sin(3\alpha) =$$
$$= \frac{1}{2\sqrt{L}}\left[3\cos(3\alpha) - \cot(\alpha)\sin(3\alpha)\right] =$$
$$= \frac{1}{2\sqrt{L}}\left[3\left(4\cos^3(\alpha) - 3\cos(\alpha)\right) - \cot(\alpha)\left(3\sin(\alpha) - 4\sin^3(\alpha)\right)\right] =$$
$$= \frac{2}{\sqrt{L}}\cos(\alpha)\left[3\cos^2(\alpha) + \sin^2(\alpha) - 3\right] =$$
$$= \frac{4}{\sqrt{L}}\cos(\alpha)(\cos^2(\alpha) - 1) = -\frac{2}{\sqrt{L}}\sin\left(\frac{\pi x}{L}\right)\sin\left(\frac{2\pi x}{L}\right)$$

Bibliography

1. Cooper, F., Khare, A. & Sukhatme, U. Supersymmetry and Quantum Mechanics. *Physics Reports* **251.** arXiv:hep-th/9405029, 267–385. ISSN: 03701573. doi:10.1016/0370-1573(94)00080-M (Jan. 1995).
2. Khare, A. *Supersymmetry in Quantum Mechanics* in *AIP Conference Proceedings* **744.** arXiv:math-ph/0409003 (2004), 133–165. doi:10.1063/1.1853201.
3. Bastianelli, F. *Meccanica Quantistica Supersimmetrica: Appunti per il corso di Fisica Teorica* 2011. https://www-th.bo.infn.it/people/bastianelli/Susyqm-12.pdf.
4. Zwiebach, B. *Mastering quantum mechanics: essentials, theory, and applications* ISBN: 9780262046138 (The MIT Press, Cambridge, MA, 2022).
5. Dutt, R., Khare, A. & Sukhatme, U. Supersymmetry-inspired WKB approximation in quantum mechanics. en. *American Journal of Physics* **59,** 723–727. ISSN: 0002-9505, 1943-2909. doi:10.1119/1.16840 (Aug. 1991).
6. Dutt, R., Gangopadhyaya, A., Khare, A., Pagnamenta, A. & Sukhatme, U. Semiclassical approach to quantum-mechanical problems with broken supersymmetry. en. *Physical Review A* **48,** 1845–1853. ISSN: 1050-2947, 1094-1622. doi:10.1103/PhysRevA.48.1845 (Sept. 1993).